

Tab 1

1. Project Information

Project Name: mock interview AI

Project Description:

Will prepare me for an interview by asking questions. As a human I will answer and it will evaluate the answer on a rating out of 10. Give me strengths and weaknesses and also the correct answer.

Project Goal:

Ask technical HR questions. A technical Java, Python, database, SQL-based question typically asked in a big organization for placement

Expected Completion Date:

7 to 15 days' time, it will help me to prepare.

Project Owner: Dr.Amar Panchal

2. AI Role

Primary Role of AI:

Technical interviewer, HR interviewer, behavioral coach, or general interview preparation coach

Additional Roles:

Placement-related question generator, behavioral coach, ATS score Evaluator, personality coach

Target Audience:

Final-year student, graduate, preparing for placement, about to go for placement, age group 22 to 29

3. Inputs

Documents:

text,pdf,docx,image of jd

Websites:

- LinkedIn
- Indeed
- NoKri.com
- Geeks for Geek
- HackerRank
- CodeShape
- LeadCode

Images:

- JD screenshot of company details
- screenshot of whiteboard or handwritten notes

4. Expected Outputs

Reports complete report about:

- my resume
- strengths
- weaknesses
- communication
- evaluation of communication
- evaluation of resume versus JD
- knowledge gap, if any gap, etc.

PDF Job description or PDF of invitation of job application

Code code blocks for finding output or explaining how the execution works or for error trapping

Flowchart start with a basic question, go on to a difficult advanced linear growing graph from simple to difficult.

5. Workflow

Will start with my resume. Either ask me a question directly on the resume or give me topics. I could choose one, start asking a question in a linear to advanced manner, one question at a time. Ask me a question, hear out my answer, and then evaluate the answer. If correction is needed, give me a proper correction.

6. Writing & Design Preferences

Language:english

Tone: motivating, clear on dot, giving plus and minus of my answer, and friendly

Expected Output:

Proper evaluation in simplest English so that I can understand. If needed, might ask for an explanation in other languages like Hindi or Marathi.

8. Constraints

One question at a time, from simple to difficult. First get the answer, give me evaluation, and then only ask next.

9. Quality Checklist

10. Success Criteria

Get to practice the right question asked in real interviews. Let it be technical, HR, or behavioral.

Key Performance Indicators (KPIs):

Out of ten strengths, weaknesses, what was wrong in my answer? What is right in my answer?

Tab 2

Option 1: AI English Interview Communication Mentor

Problem Definition

Many students understand English grammar and vocabulary but fail to communicate confidently during interviews. They struggle with speaking fluently, forming professional sentences, answering HR and technical interview questions, improving pronunciation, and maintaining confidence while speaking. Traditional English learning focuses on grammar instead of interview-oriented communication.

There is a need for an AI mentor that acts like a personal English communication coach. The AI should assess the student's current communication level, identify weak areas, create personalized practice sessions, conduct mock interview conversations, evaluate every response, suggest better alternatives, track improvement over time, and continuously motivate the learner until they become interview-ready.

The system should focus on practical communication rather than theoretical English and simulate real interview environments.

Target Users

- Freshers
- Engineering students
- College students
- Job seekers
- Professionals preparing for interviews

Expected Outcome

Students become confident in speaking English during interviews through personalized practice, instant feedback, and continuous improvement.

Option 2: AI Smart Study Planner

Problem Definition

Students often know what they need to study but do not know how to plan their preparation effectively. They either spend too much time on easy topics or insufficient time on difficult ones. Many fail to complete their syllabus because they cannot estimate the effort required for each topic or create a realistic study schedule.

There is a need for an AI mentor that intelligently creates personalized study plans based on syllabus difficulty, available study time, deadlines, learning speed, and revision requirements.

The AI should first gather information such as:

- Subject
- Topics
- Difficulty level
- Examination date
- Daily available study hours
- Current preparation level
- Preferred study timings

Based on this information, it should generate a complete day-wise study roadmap including learning sessions, revision sessions, practice problems, mock tests, buffer days, and progress tracking. The planner should also adjust the schedule whenever the student misses a day or changes availability.

Target Users

- School students
- College students
- Competitive exam aspirants
- Placement preparation students
- Self-learners

Expected Outcome

Students follow a realistic, personalized study plan that maximizes productivity, improves consistency, and increases the likelihood of completing the syllabus on time.

Option 3: AI Psychometric Career Mentor

Problem Definition

Many students choose careers based on trends, family influence, or peer pressure without understanding their own personality, strengths, weaknesses, interests, and work preferences. This often leads to poor career decisions, low confidence, and dissatisfaction.

There is a need for an AI mentor that conducts a structured psychometric assessment through interactive questions and scenario-based exercises. The AI should analyze the responses to identify personality traits, cognitive abilities, communication style, leadership potential, learning preferences, emotional intelligence, problem-solving skills, and work behavior.

Based on the assessment, the AI should generate a detailed report explaining the student's strengths, weaknesses, suitable career paths, areas for improvement, learning style, interview readiness, and personalized recommendations for skill development.

The mentor should also suggest practical activities and resources to improve weaker areas and periodically reassess progress.

Target Users

- School students
- College students
- Fresh graduates
- Career changers
- Placement candidates

Expected Outcome

Students gain a deeper understanding of themselves, make informed career decisions, improve weaker skills, and align their learning with careers that match their personality and strengths.

Option 4: AI Quantitative Aptitude Mentor for Weak Mathematics Students

Problem Definition

Many students from diploma, non-technical, rural, or weak mathematics backgrounds struggle to prepare for quantitative aptitude required in placement tests and competitive examinations. They often have gaps in fundamental concepts such as percentages, fractions, ratios, algebra, speed-distance-time, probability, and logical calculations. As a result, they lose confidence, avoid solving problems, and perform poorly in aptitude rounds despite having good technical skills.

There is a need for an AI mentor that teaches quantitative aptitude from first principles instead of assuming prior mathematical knowledge. The AI should diagnose the student's current skill level through an initial assessment, identify conceptual gaps, and create a personalized learning path that gradually builds confidence and problem-solving ability.

The mentor should explain every concept using simple language, real-life examples, visual reasoning where applicable, and step-by-step solutions. It should adapt the difficulty based on the student's performance, provide unlimited practice questions, analyze mistakes, and revise weak topics until mastery is achieved.

The AI should also prepare students specifically for placement examinations by conducting topic-wise quizzes, timed practice sessions, mock aptitude tests, interview-style quantitative questions, and progress tracking. Every incorrect answer should become a learning opportunity through detailed explanations and alternative solving methods.

Core Features

- Initial mathematics skill assessment
- Identification of strengths and weak areas
- Personalized learning roadmap
- Beginner-friendly explanations using real-life examples
- Step-by-step problem-solving approach
- Multiple solving techniques (traditional, shortcut, and logical methods)
- Daily practice questions based on previous performance
- Adaptive difficulty level
- Topic-wise quizzes and mock placement tests
- Speed improvement through timed practice
- Performance analytics and progress reports
- Revision plans for weak topics
- Motivation and confidence-building feedback
- Interview-oriented quantitative aptitude preparation

Target Users

- Diploma students
- Polytechnic students
- Engineering students with weak mathematics
- Rural-background learners
- Freshers preparing for placements
- Competitive examination aspirants
- Students restarting mathematics after a long gap

Expected Outcome

Students who begin with weak mathematical fundamentals develop strong quantitative aptitude skills through structured, personalized learning. By the end of the program, they should be able to confidently solve placement-level aptitude questions with improved accuracy, speed, and logical thinking, making them well-prepared for campus placements, company aptitude tests, and technical interviews.